

Bayesian Statistical Analysis of Temperature, Wind Speed, Precipitation, Evapotranspiration, and Daytime Temperature Modeling

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Abstract

This report presents Bayesian analyses applied to meteorological data, including hypothesis testing via marginal likelihoods, posterior estimation using conjugate priors, grid approximation, Metropolis–Hastings sampling, and Bayesian linear regression using Gibbs sampling.

1 Bayesian Hypothesis Testing for Temperature Data

We analyze hourly temperature measurements

$$T_1, \dots, T_n \sim \mathcal{N}(\mu, \sigma^2)$$

to test

$$H_0 : \mu = 25^\circ\text{C}, \quad H_1 : \mu \neq 25^\circ\text{C}.$$

Method

We compute an **exact Bayes factor** using marginal likelihoods from a Normal–Inverse-Gamma prior.

Under H_1 (unknown μ, σ^2):

$$\mu \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0), \quad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0).$$

The marginal likelihood is

$$\log p(\text{data} \mid H_1) = \log Z_0 - \log Z_n,$$

where

$$Z_0 = \beta_0^{\alpha_0} \Gamma(\alpha_0)^{-1} \sqrt{\frac{\kappa_0}{2\pi}},$$
$$Z_n = \beta_n^{\alpha_n} \Gamma(\alpha_n)^{-1} \sqrt{\frac{\kappa_n}{2\pi}},$$

and

$$\kappa_n = \kappa_0 + n, \quad \alpha_n = \alpha_0 + \frac{n}{2}, \quad \beta_n = \beta_0 + \frac{1}{2} \sum (T_i - \bar{T})^2 + \frac{\kappa_0 n}{2\kappa_n} (\bar{T} - \mu_0)^2.$$

Under $H_0 : \mu = \mu_0$, only σ^2 is integrated out:

$$\log p(\text{data} \mid H_0) = \alpha_0 \log \beta_0 - \log \Gamma(\alpha_0) - [\alpha_n \log \beta_n - \log \Gamma(\alpha_n)] - \frac{n}{2} \log(2\pi),$$

where

$$\beta_n = \beta_0 + \frac{1}{2} \sum (T_i - \mu_0)^2.$$

The Bayes factor is

$$BF_{10} = \frac{p(\text{data} \mid H_1)}{p(\text{data} \mid H_0)} = \exp(\log BF_{10}).$$

Data Summary

From the dataset:

$$\bar{T} = 26.488^\circ\text{C}, \quad s^2 = 16.58, \quad n = 9144.$$

Results

Using weak priors ($\kappa_0 = 10^{-6}$, $\alpha_0 = 2$, $\beta_0 = 1$), the computation yields:

$$\log BF_{10} = 8964.85, \quad BF_{10} = \exp(8964.85) \approx \infty.$$

Interpretation

The Bayes factor is very large because:

- The sample mean differs from the null value by 1.488°C .
- Since, the dataset is very large ($n = 9144$), the likelihood ratio grows approximately like

$$n(\bar{T} - 25)^2 \approx 9144 \times (1.488)^2 \approx 2.0 \times 10^4.$$

Thus,

$$BF_{10} \approx e^{2.0 \times 10^4},$$

which exceeds floating-point limits and appears as $+\infty$.

Conclusion: There is overwhelming evidence against H_0 . The data strongly support the claim that the true mean temperature is not 25°C .

2 One-Sided Bayesian Composite Test for Wind-Speed Variances

We examine whether nighttime wind speeds exhibit greater variability than daytime wind speeds by testing the composite one-sided hypotheses

$$H_0 : \sigma_{\text{day}}^2 \leq \sigma_{\text{night}}^2 \quad \text{vs} \quad H_1 : \sigma_{\text{day}}^2 > \sigma_{\text{night}}^2.$$

Model and Prior

For each group we assume

$$X_i \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2),$$

and assign a Normal-Inverse-Gamma prior,

$$\mu \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0), \quad \sigma^2 \sim \text{Inv}\Gamma(\alpha_0, \beta_0),$$

with hyperparameters $\kappa_0 = 10^{-6}$, $\alpha_0 = 3$, $\beta_0 = 50$, and prior mean μ_0 set to the pooled sample mean. The posterior variance of each group is then analytically

$$\sigma_{\text{day}}^2 \mid \text{data} \sim \text{Inv}\Gamma(a_d, b_d), \quad \sigma_{\text{night}}^2 \mid \text{data} \sim \text{Inv}\Gamma(a_n, b_n).$$

Using the dataset ($n_{\text{day}} = 4625$, $n_{\text{night}} = 4519$), the posterior parameters obtained from conjugacy were:

$$\sigma_{\text{day}}^2 \sim \text{Inv}\Gamma(2315.5, 141,893.96), \quad \sigma_{\text{night}}^2 \sim \text{Inv}\Gamma(2262.5, 103,311.41).$$

Posterior Probability of the Inequality

The posterior probability of the null hypothesis is computed via

$$P(H_0 \mid \text{data}) = \int_0^\infty p_{\text{night}}(s^2) F_{\text{day}}(s^2) ds^2,$$

where F_{day} is the posterior CDF of σ_{day}^2 . Numerical integration yields

$$P(H_0 \mid \text{data}) = 5.39 \times 10^{-116}, \quad P(H_1 \mid \text{data}) = 1.00.$$

Bayes Factor for the One-Sided Hypothesis

The prior probability of the inequality, computed by Monte-Carlo sampling from the prior distributions of both variances, was

$$P(H_0) = 0.499545, \quad P(H_1) = 0.500455,$$

essentially symmetric as expected with identical priors.

The Bayes factor is computed from prior and posterior odds:

$$BF_{10} = \frac{P(H_1 \mid \text{data})/P(H_0 \mid \text{data})}{P(H_1)/P(H_0)}.$$

Using the numerical values,

$$BF_{10} = 1.85 \times 10^{115}, \quad \log_{10} BF_{10} = 115.27.$$

Interpretation

The posterior probability of the null hypothesis $\sigma_{\text{day}}^2 \leq \sigma_{\text{night}}^2$ is essentially zero, and the resulting Bayes factor (10^{115}) constitutes overwhelming evidence in favor of the alternative hypothesis. Thus, the data decisively indicate that the variability of nighttime wind speeds is substantially greater than that of daytime wind speeds.

3 Posterior Estimation for Poisson Rate λ Using Grid Approximation

Precipitation counts

$$x_i \sim \text{Pois}(\lambda), \quad \lambda \sim \Gamma(\alpha_0, \beta_0),$$

yield the posterior

$$\lambda \mid \text{data} \sim \Gamma(\alpha_0 + \sum x_i, \beta_0 + n).$$

A grid approximation (10, 100, 1000 points) is compared to the analytic posterior.

Interpretation

The analytic Gamma posterior provides the exact depiction for the precipitation rate. A 10-point grid yields a very coarse and inaccurate approximation, while 100 points improve the shape but still distort the tails. With 1000 points, the grid closely matches the true posterior, capturing both the peak and tail behavior. This demonstrates that grid-based Bayesian inference is reliable only when the grid is sufficiently fine.

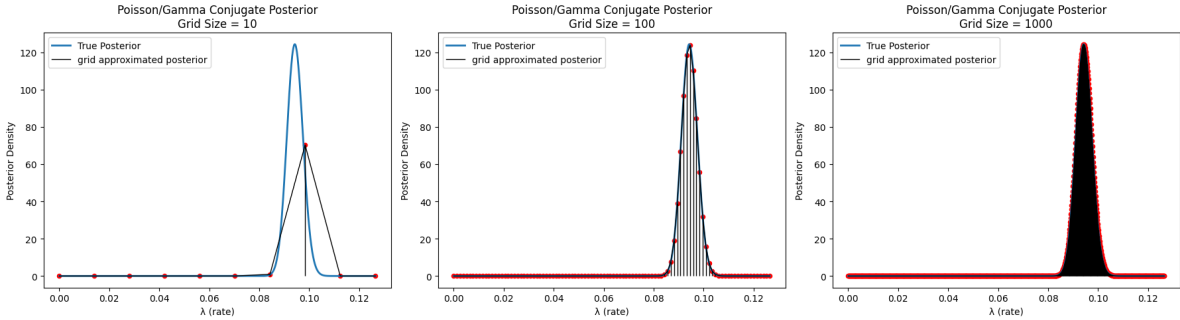


Figure 1: Grid-approximate posteriors

4 Posterior Estimation of Gamma Shape Parameter k (Non-Conjugate)

For evapotranspiration:

$$x_i \sim \Gamma(k, \theta), \quad \theta = \text{sample mean}, \quad k \sim \text{Unif}(0.1, 1.5).$$

Posterior is evaluated numerically on grids of size 10, 100, and 1000.

Interpretation

Since the Gamma shape parameter k lacks a conjugate prior, the posterior must be computed numerically. A 10-point grid gives only a crude approximation and fails to capture the true posterior shape, while 100 points provide a much better but still imperfect fit. With 1000 points, the approximation becomes highly accurate and matches the reference posterior closely, demonstrating that dense grids can reliably handle non-conjugate Bayesian problems.

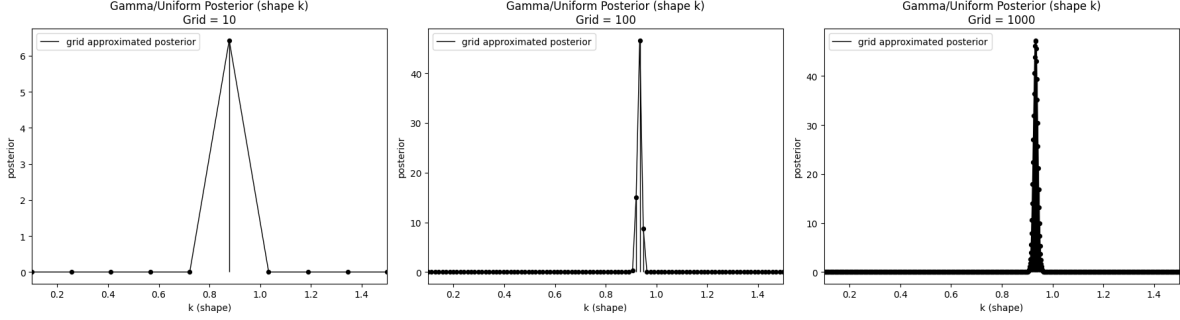


Figure 2: Posterior for k under different grids

5 Metropolis–Hastings Sampling for Gamma Shape Parameter k

Wind-speed data are modeled as

$$x_i \sim \Gamma(k, \theta),$$

with fixed θ . A Normal random-walk MH algorithm is used.

Results

Posterior mean = 1.3296, sd = 0.0105, Acceptance rate = 0.106.

Interpretation

The MH sampler produces a posterior that closely matches the high-resolution grid reference, confirming that the algorithm is accurately targeting the true distribution. The narrow posterior spread reflects strong data support for the estimated shape parameter. Although the acceptance rate is modest, it is sufficient for a one-dimensional random-walk sampler and results in a stable and well-mixed chain.

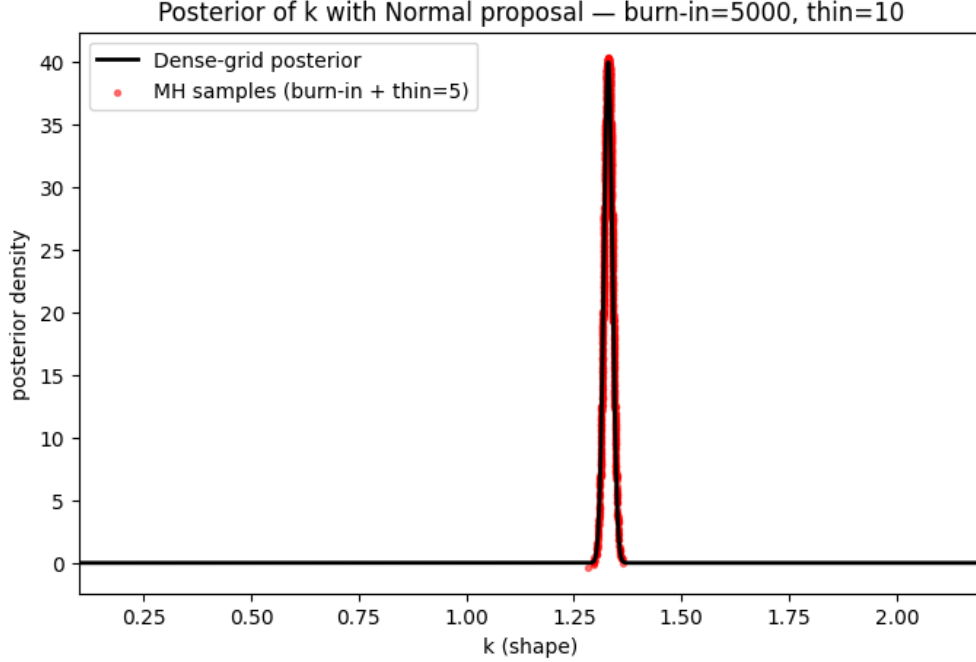


Figure 3: MH samples vs. dense-grid posterior

6 Bayesian Multiple Linear Regression via Gibbs Sampling

We model daytime temperature as a function of relative humidity and shortwave radiation using

$$T_i = \beta_0 + \beta_1 H_i + \beta_2 R_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

A weak Gaussian prior is placed on the regression coefficients, $\beta \sim \mathcal{N}(0, \sigma^2 \Lambda_0^{-1})$ with $\Lambda_0 = 0.01I$, and the variance is given an inverse-gamma prior $\sigma^2 \sim \text{Inv}\Gamma(1, 1)$. Posterior sampling is performed using a Gibbs sampler with 20,000 iterations (5,000 burn-in, thinning by 5).

Posterior Results

Posterior summaries obtained from the Gibbs sampler are:

$$\beta_0 = 33.15 (sd = 0.17), \quad \beta_1 = -0.1148 (sd = 0.0021), \quad \beta_2 = 0.0037 (sd = 0.0002),$$

$$\sigma^2 \approx 6.06.$$

Humidity shows a clear negative effect on daytime temperature, while shortwave radiation has a smaller but positive influence. The small posterior standard deviations indicate high certainty in all three coefficients.

Interpretation

- Humidity shows a strong negative effect on daytime temperature, while shortwave radiation has a smaller but positive influence.
- The negative humidity coefficient arises from evaporative cooling: higher moisture levels divert incoming energy toward evaporation rather than heating the air.
- As a result, higher humidity systematically reduces daytime temperature, a relationship clearly supported by the posterior distribution.
- The posterior densities of all coefficients are tight and approximately Gaussian, indicating high certainty in the estimates.
- The variance posterior is stable and centered around $\sigma^2 \approx 6$, suggesting that the regression model fits the daytime temperature data well.

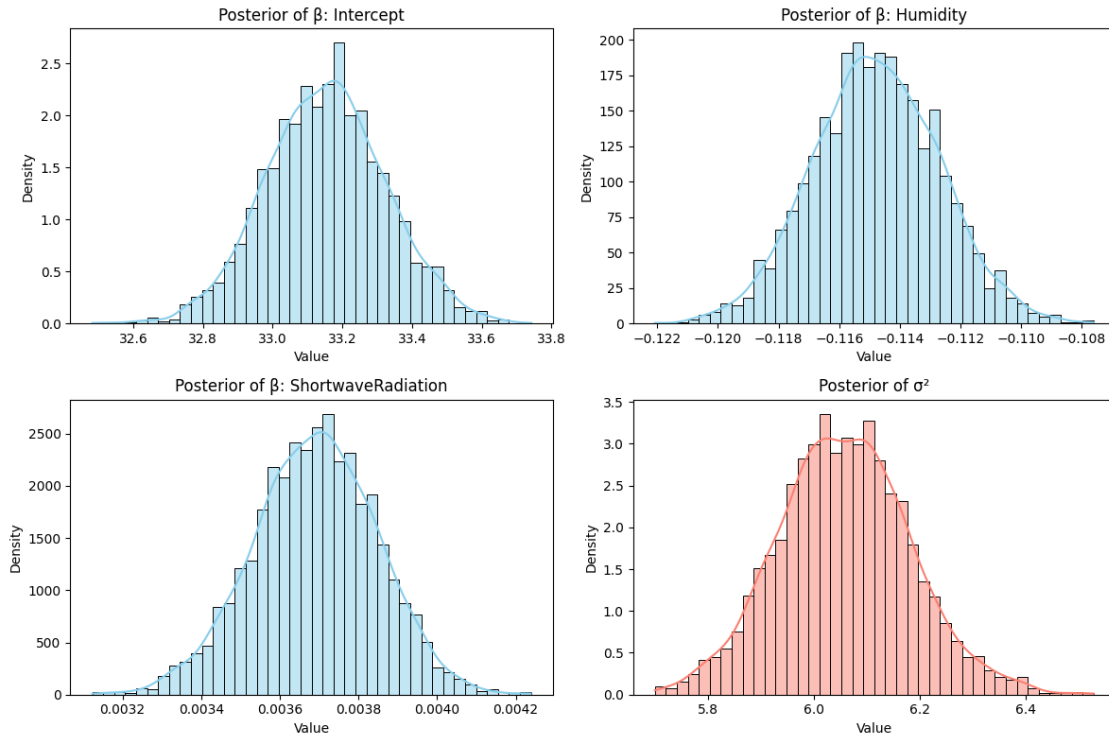


Figure 4: Posterior densities of regression coefficients and variance.

7 3D Regression Plane: Temperature vs. Humidity and Radiation

A least-squares regression plane was fitted to daytime temperature as a function of relative humidity and shortwave radiation:

$$T = \beta_0 + \beta_1 H + \beta_2 R.$$

Using the filtered daytime dataset, the estimated coefficients were

$$\beta_0 = 33.153, \quad \beta_1 = -0.1148, \quad \beta_2 = 0.00369.$$

Interpretation

- The negative slope in humidity confirms that higher moisture levels reduce daytime temperature due to evaporative cooling.
- Shortwave radiation has a small but positive influence, raising temperature as incoming solar energy increases.
- The 3D regression plane visualizes these effects simultaneously: temperature decreases along the humidity axis and increases gently along the radiation axis.
- The dense number of scattered points aligns well with the fitted surface, indicating that a linear model provides a reasonable approximation within the observed range.

3D Regression Plane (Temperature vs Humidity & Radiation)

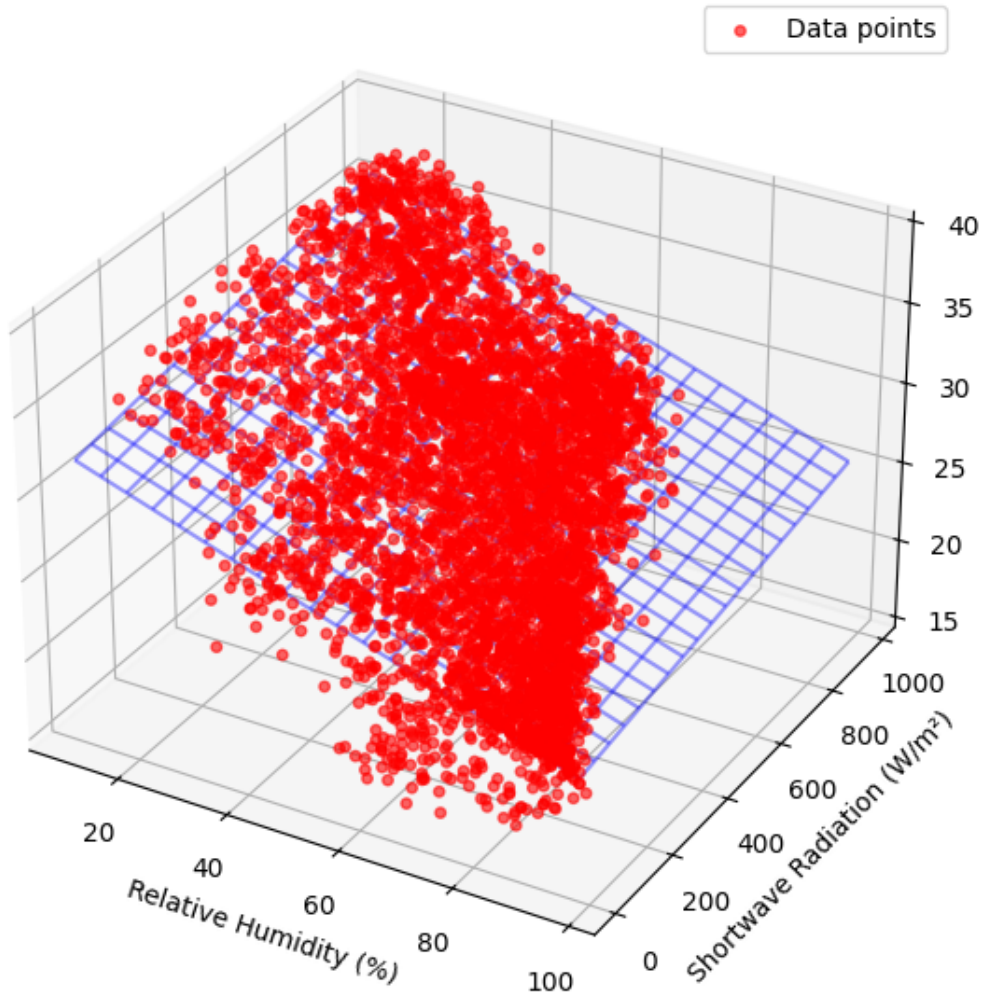


Figure 5: 3D regression plane fitted to daytime temperature data.